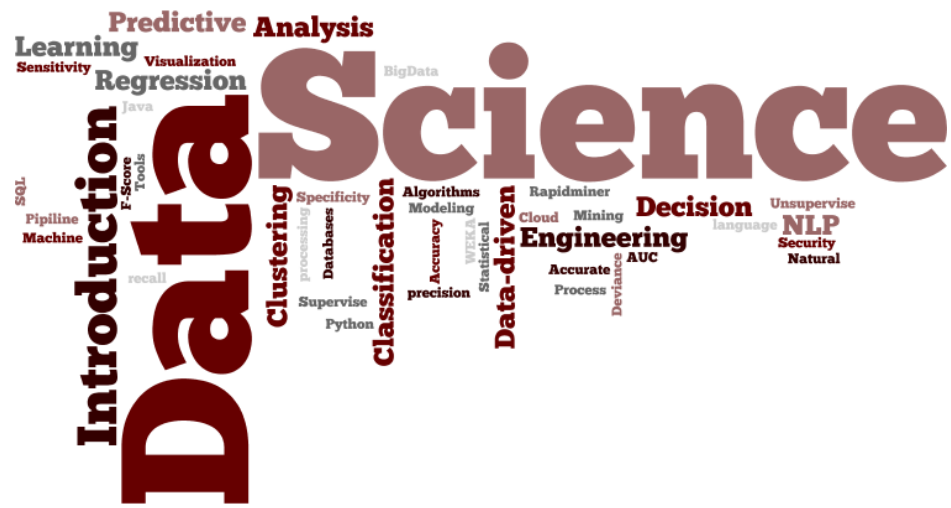




9.2 เครื่องมือวิทยาการข้อมูล: Rapidminer

Tools for Data Science: Rapidminer – Regression3

ดำรงฤทธิ์ เศรษฐศิริโชค

Regression LAB 3

- Case Study
Water Pump RUL - Predictive Maintenance



Regression

LAB 3

Water Pump RUL - Predictive Maintenance



1.Business Understanding

- Water Pump RUL

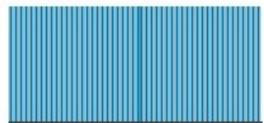
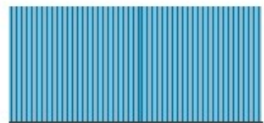
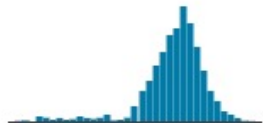
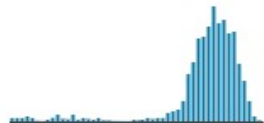
Use Case:

<https://www.kaggle.com/datasets/anseldsouza/water-pump-rul-predictive-maintenance>

Original:

<https://www.kaggle.com/datasets/nphantawee/pump-sensor-data>

2.Data Understanding

This file contains 3 main group o of data 1. Timestamp data 2. Sensor data(52 series): All values are raw values 3. The remaining useful life before the next failure in hours						
#	timestamp	# sensor_00	# sensor_01	# sensor_02	# sensc	
 0 166k	 1Apr18 25Jul18	 0 2.55 22.4 55.8	 33.2 56	 31.6		
0	2018-04-01 00:00:00	2.465394	47.09201	53.2118	46.3107	
1	2018-04-01 00:01:00	2.465394	47.09201	53.2118	46.3107	
2	2018-04-01 00:02:00	2.444734	47.35243	53.2118	46.3971	
3	2018-04-01 00:03:00	2.460474	47.09201	53.1684	46.3971	
4	2018-04-01 00:04:00	2.445718	47.13541	53.2118	46.3971	
5	2018-04-01 00:05:00	2.453588	47.09201	53.1684	46.3971	
6	2018-04-01 00:06:00	2.455556	47.04861	53.168399810790994	46.3971	
7	2018-04-01 00:07:00	2.440550	47.10511	53.168399810790994	46.3971	

2.Data Understanding

Views: []

Result History: ExampleSet (Retrieve rul_hrs) [X]

Open in: Turbo Prep | Auto Model

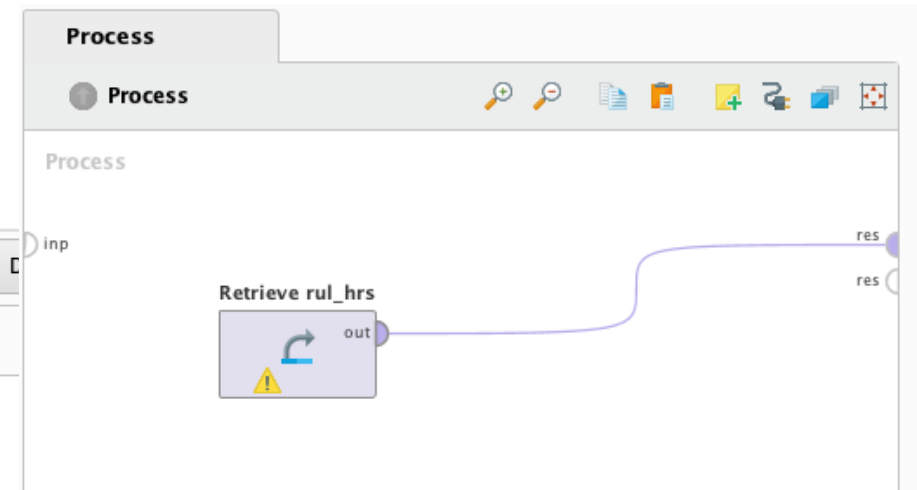
Data

Statistics

Visualizations

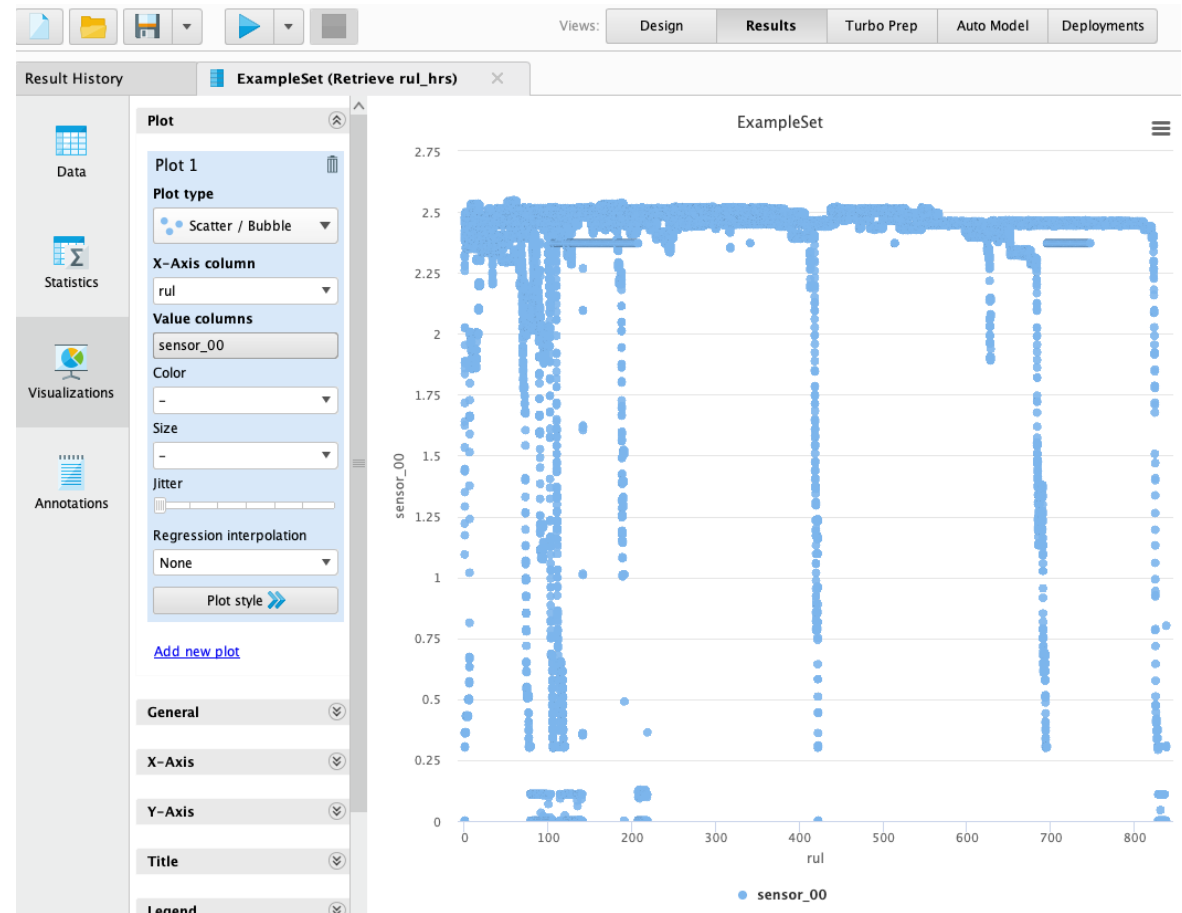
Annotations

Row No.	att1	timestamp	sensor_00	sensor_01	sensor_02	sensor_03	sen
1	0	Apr 1, 201...	2.465	47.092	53.212	46.311	634
2	1	Apr 1, 201...	2.465	47.092	53.212	46.311	634
3	2	Apr 1, 201...	2.445	47.352	53.212	46.398	638
4	3	Apr 1, 201...	2.460	47.092	53.168	46.398	628
5	4	Apr 1, 201...	2.446	47.135	53.212	46.398	636
6	5	Apr 1, 201...	2.454	47.092	53.168	46.398	637
7	6	Apr 1, 201...	2.456	47.049	53.168	46.398	633
8	7	Apr 1, 201...	2.450	47.135	53.168	46.398	630
9	8	Apr 1, 201...	2.463	47.092	53.168	46.398	631
10	9	Apr 1, 201...	2.446	47.179	53.168	46.398	641
11	10	Apr 1, 201...	2.464	47.483	53.125	46.398	637
...	...	...	...	...	...	...	...



2.Data Understanding

- ทดลอง plot สังเกตว่า
ไม่สามารถบ่งบอก
อะไรได้เท่าที่ควร



ทดลอง Filter

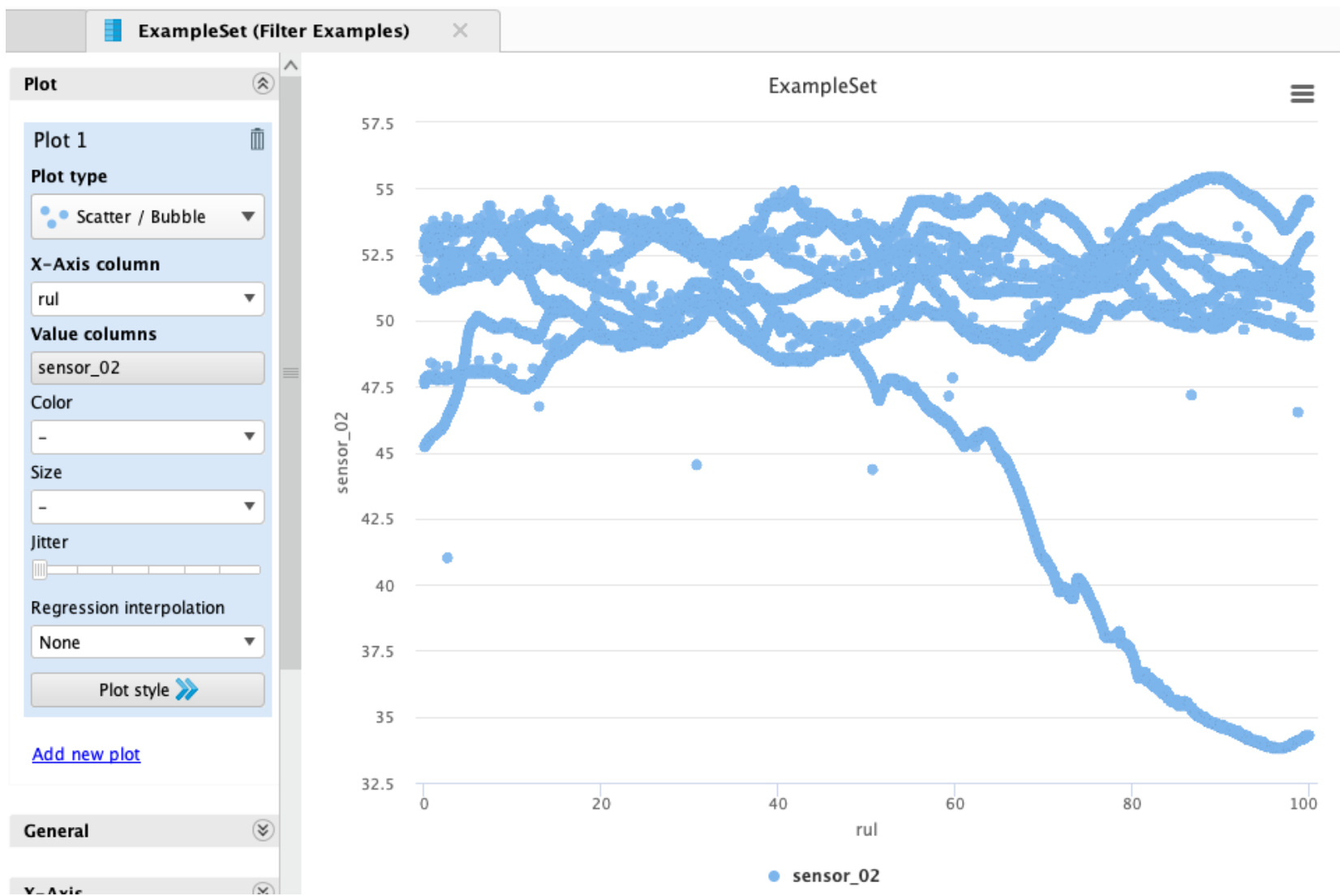
The image shows a workflow diagram and a dialog box for creating filters.

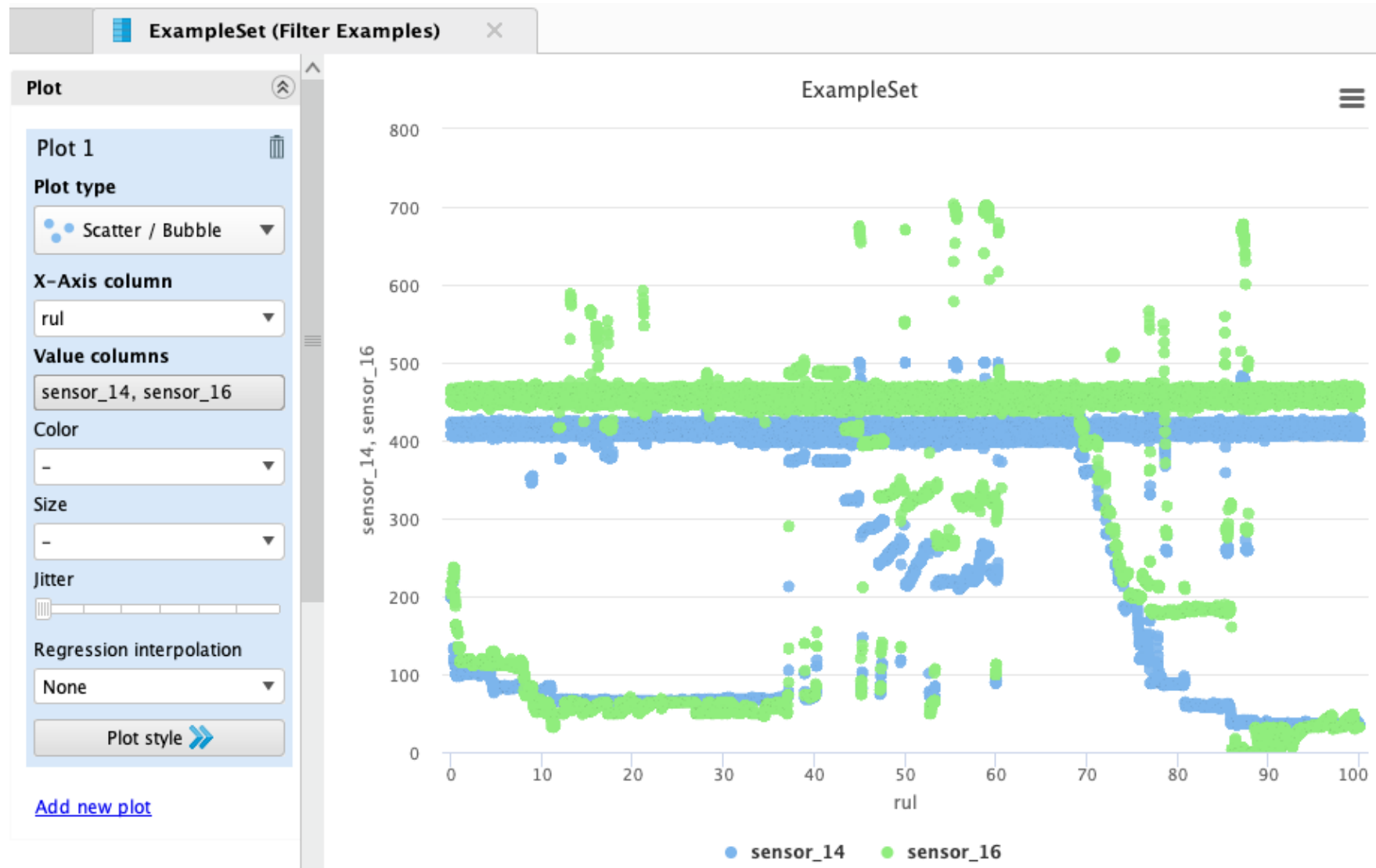
Workflow Diagram:

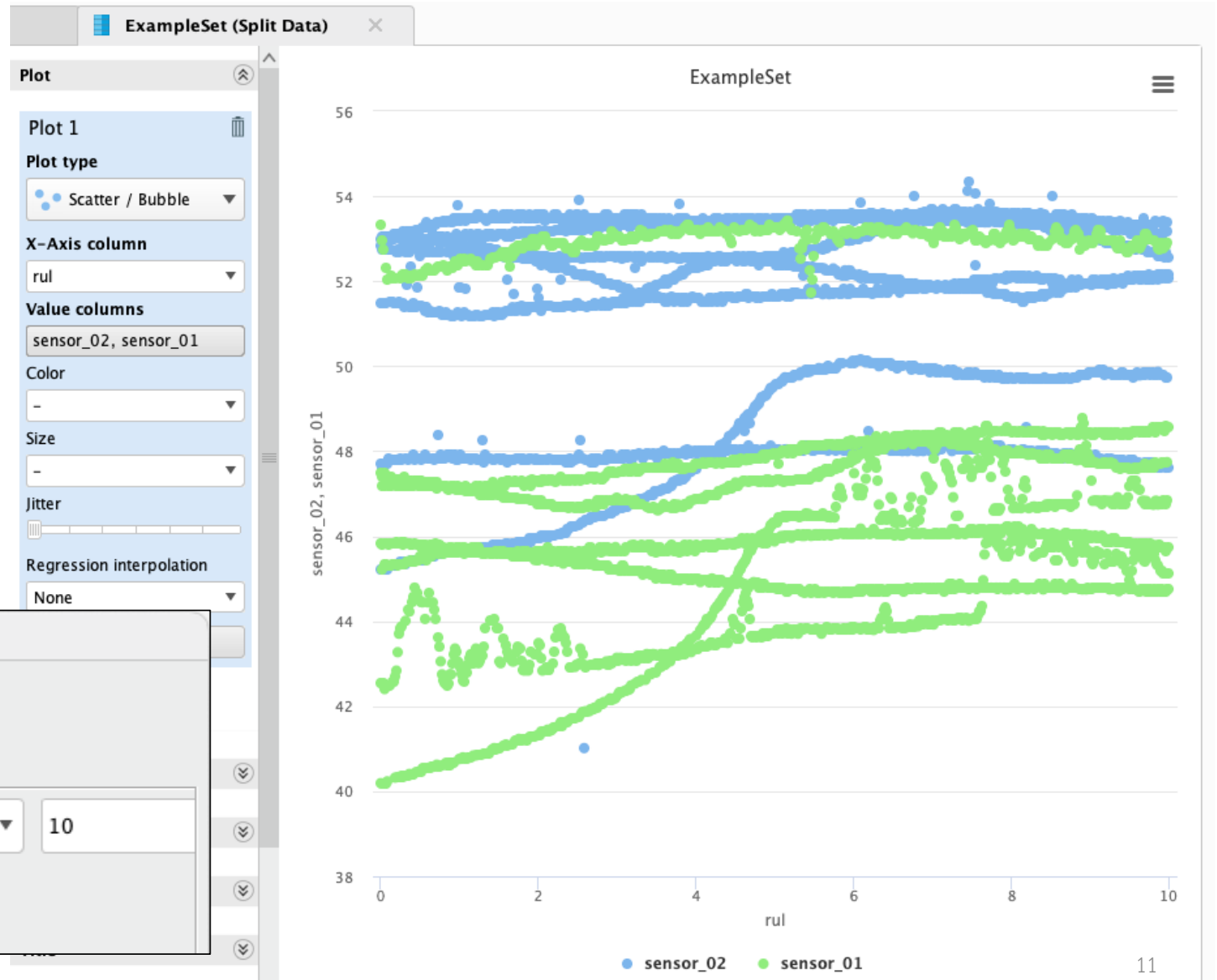
- A box labeled "Retrieve rul_hrs" with an "out" port is connected to a box labeled "Filter Examples".
- The "Filter Examples" box has four ports: "exa" (input), a funnel icon (input), "exa" (output), and "ori" (output).
- The "ori" port is connected to a box labeled "unm".
- The "unm" box is connected to a box labeled "res".

Create Filters: filters Dialog Box:

- Title: Create Filters: filters
- Subtitle: Defines the list of filters to apply.
- Input fields: "rul" (dropdown), " $\leq$ " (dropdown), and "100" (text field).
- Buttons: "Add Entry" (with a plus icon), "OK" (with a green checkmark), and "Cancel" (with a red X).



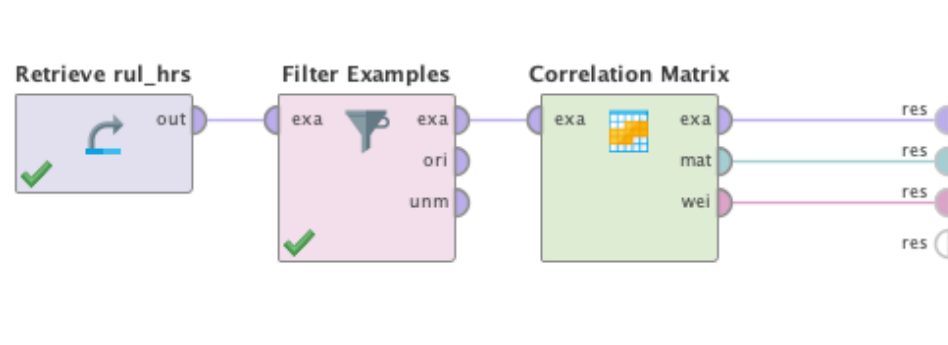




Create Filters: filters

Create Filters: **filters**
Defines the list of filters to apply.

rul ≤ 10

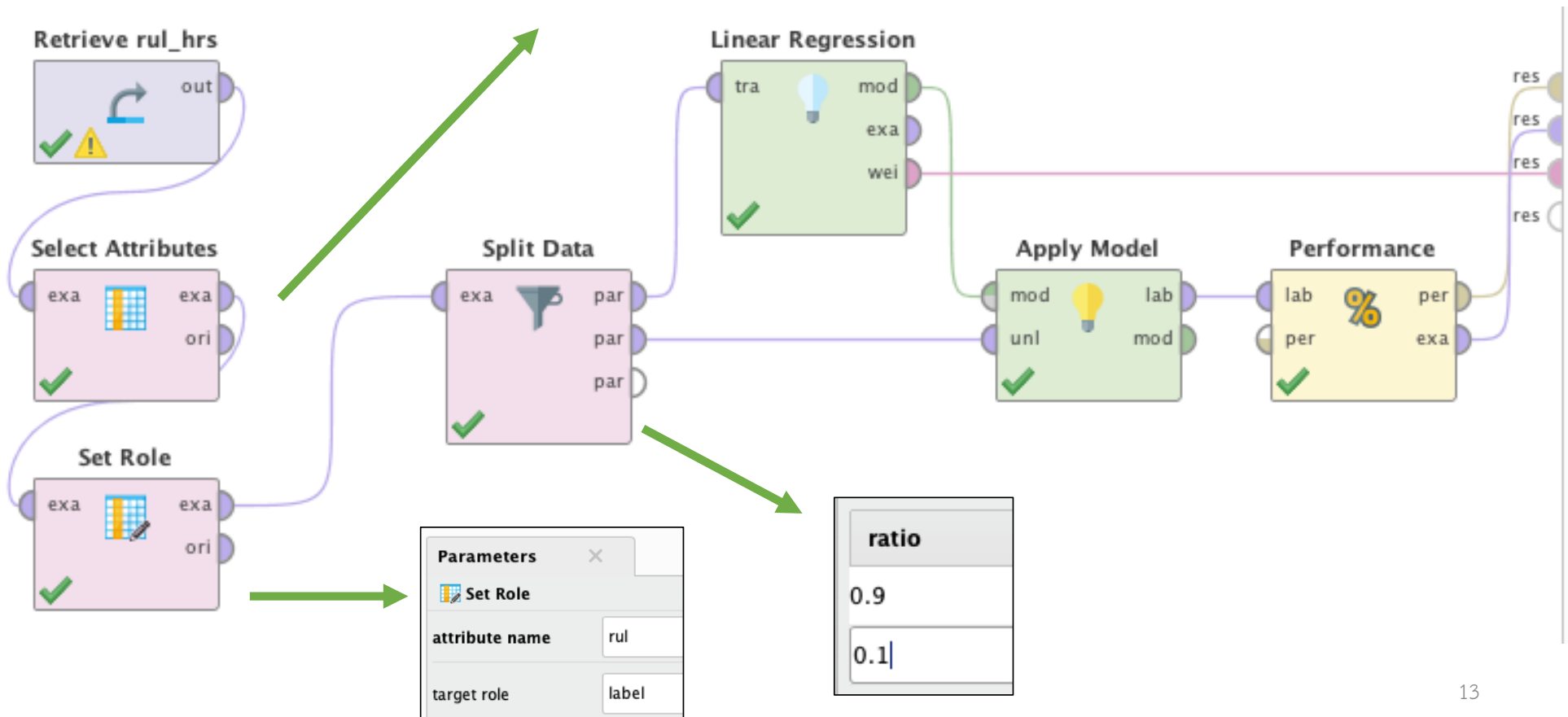


Correlation Matrix (Correlation Matrix)		ExampleSet (Filter Examples)							
Result History			AttributeWeights (Correlation Matrix)						
Attributes	att1	timestamp	sensor_00	sensor_01	sensor_02	sensor_03	sensor_04	sensor_05	sensor_06
att1	1	?	-0.219	-0.490	0.110	0.163	-0.238	-0.171	0.077
timestamp	?	1	?	?	?	?	?	?	?
sensor_00	-0.219	?	1	0.410	0.631	0.344	0.934	0.794	0.732
sensor_01	-0.490	?	0.410	1	0.649	0.686	0.346	0.418	0.036
sensor_02	0.110	?	0.631	0.649	1	0.786	0.527	0.548	0.376
sensor_03	0.163	?	0.344	0.686	0.786	1	0.269	0.322	0.169
sensor_04	-0.238	?	0.934	0.346	0.527	0.269	1	0.765	0.628
sensor_05	-0.171	?	0.794	0.418	0.548	0.322	0.765	1	0.430
sensor_06	0.077	?	0.732	0.036	0.376	0.169	0.628	0.430	1

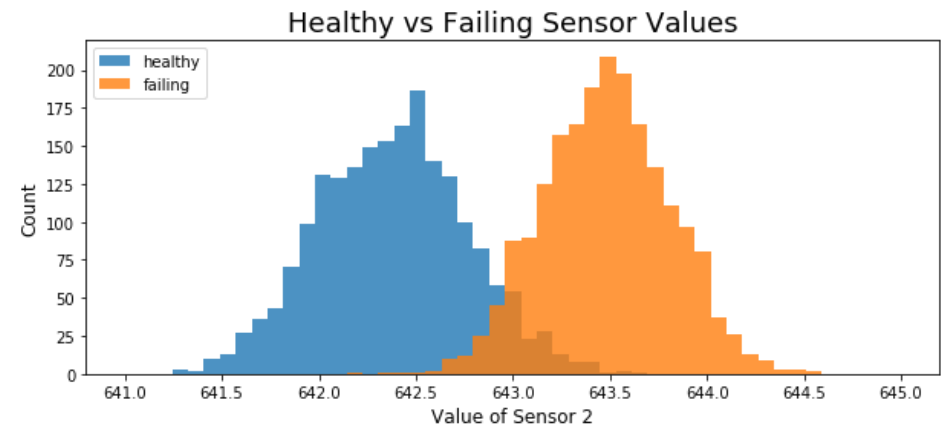
3.Data Preparation

4.Modeling

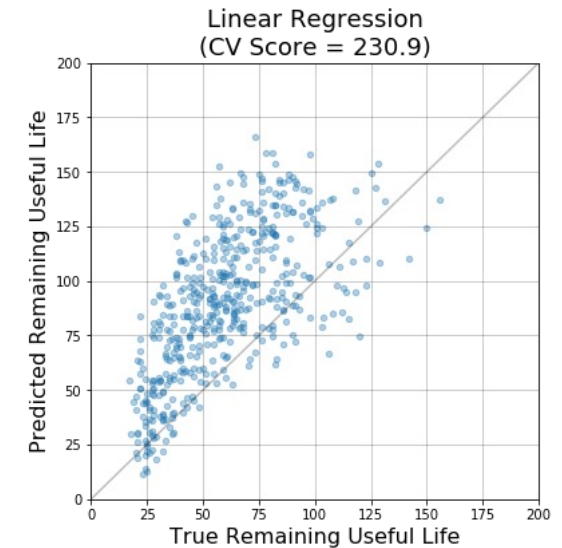
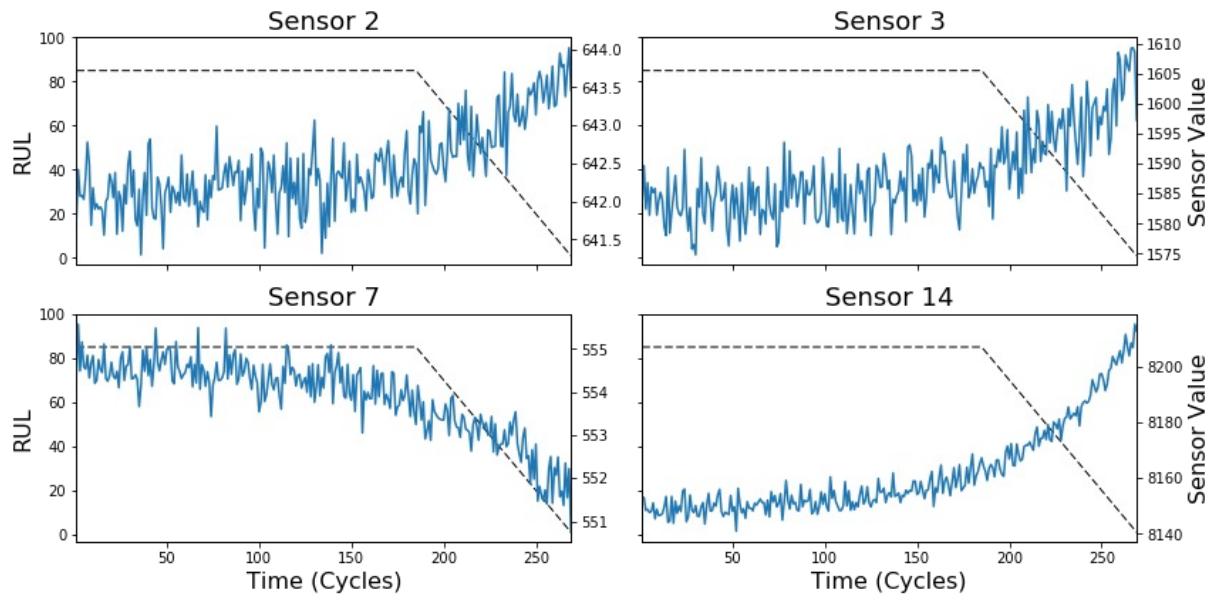
- ทดลองเลือกบาง attribute (sensor 00 - 20) RUN และสังเกตผลลัพธ์
- ทดลองใช้ทุก attribute ทดลอง RUN และสังเกตผลลัพธ์



ปรับปรุง model โดยทำความเข้าใจ ข้อมูลเพิ่มเติม



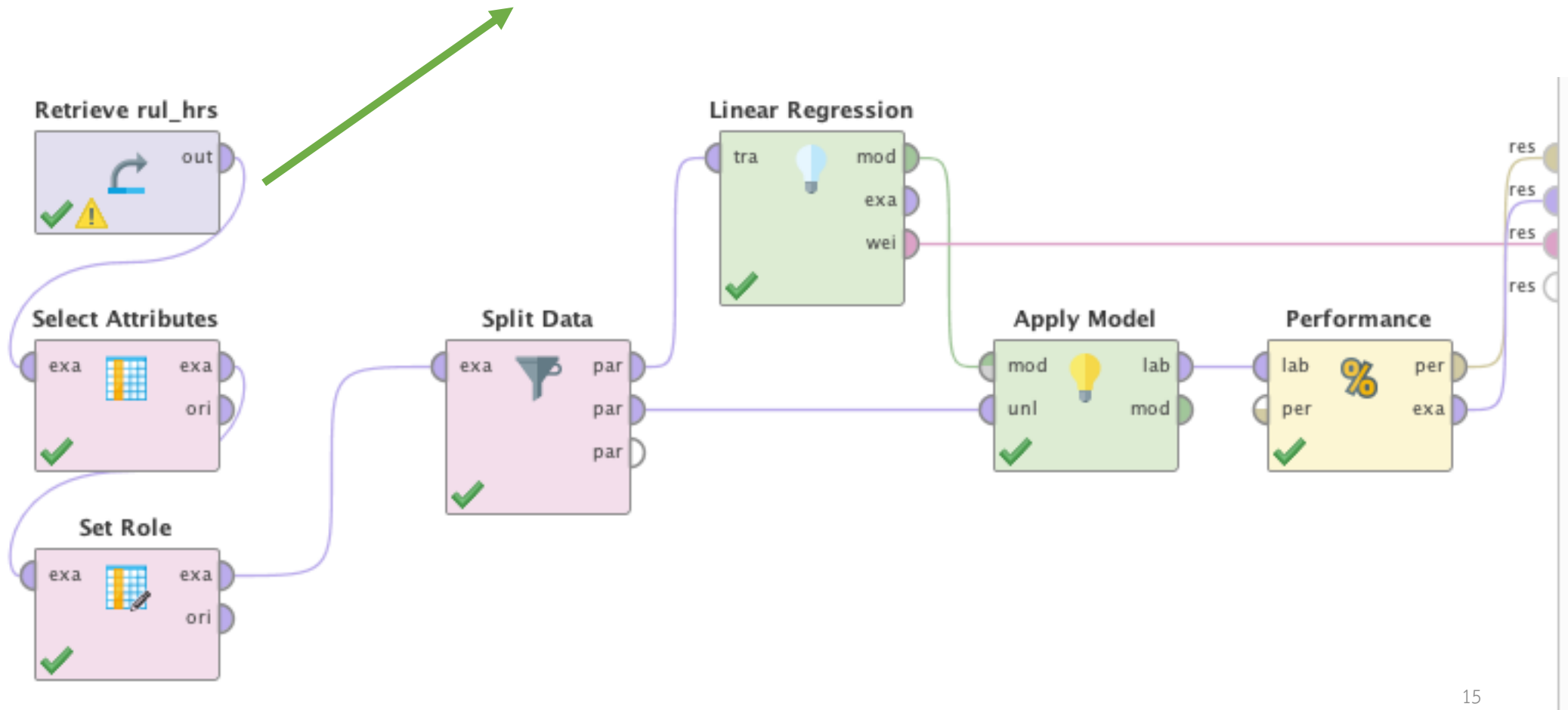
Remaining Life and Sensor Values vs Time, Example Engine



- Ref: <https://www.svds.com/getting-started-predictive-maintenance-models/>

Modeling [2]

- ใช้ข้อมูล `rul_hrs_s.csv` ทดสอบ RUN และสังเกตผลลัพธ์
- ข้อมูลตัดมาเฉพาะ **100 examples** ก่อนและหลัง machine breakdown



ตัวอย่างผลลัพธ์

Row No.	rul	prediction(...	sensor_00	se
1	285.900	270.551	2.465	47
2	285.817	266.938	2.456	47
3	285.733	287.639	2.445	47
4	285.650	233.632	2.450	48
5	285.583	300.138	2.446	49
6	285.500	301.791	2.450	49
7	285.467	246.640	2.449	48
8	285.233	327.167	2.454	49
9	285.200	312.244	2.449	49
10	285	369.087	2.464	49
11	284.950	363.946	2.446	49
12	284.883	345.660	2.459	49
13	284.667	211.484	2.464	47
14	284.650	195.800	2.446	47
15	284.500	251.097	2.442	47
16	284.417	279.616	2.446	47
17	284.333	274.884	2.459	47
18	284.283	309.459	2.459	48
19	1.650	-35.825	2.457	52
20	1.417	-61.938	2.459	52
21	1.183	-57.767	2.450	52
22	1.100	-44.292	2.449	52
23	1.033	-17.394	2.447	52

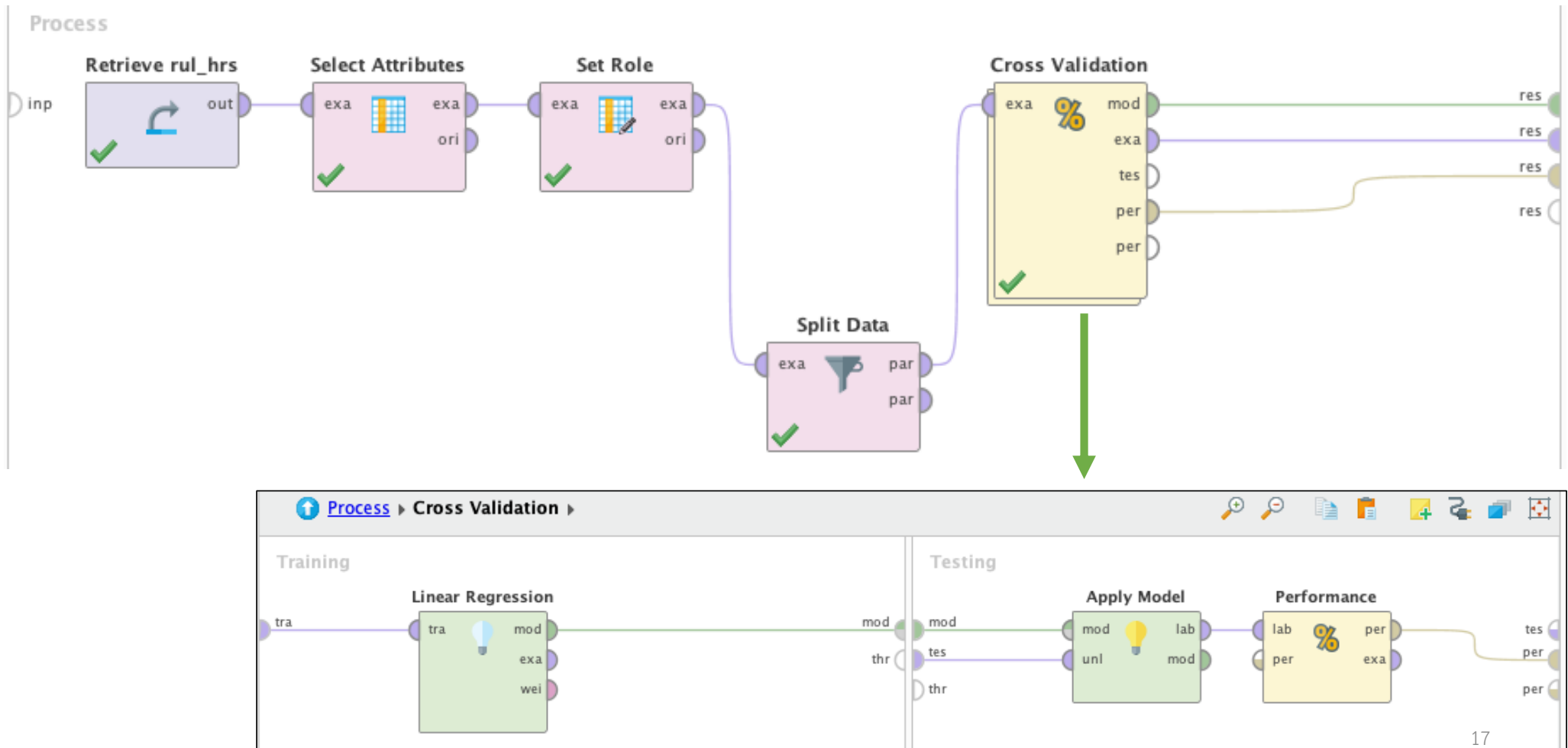
ExampleSet (281 examples, 2 special attributes, 50 regular attributes)

PerformanceVector

PerformanceVector:
 root_mean_squared_error: 92.526 +/- 0.000
 squared_error: 8560.972 +/- 19649.285
 correlation: 0.943

5.Evaluation

- ทดลอง 10-fold cross-validation สังเกตผลลัพธ์ Performance ของ model



ตัวอย่างผลลัพธ์จาก 10-fold cross-validation

PerformanceVector

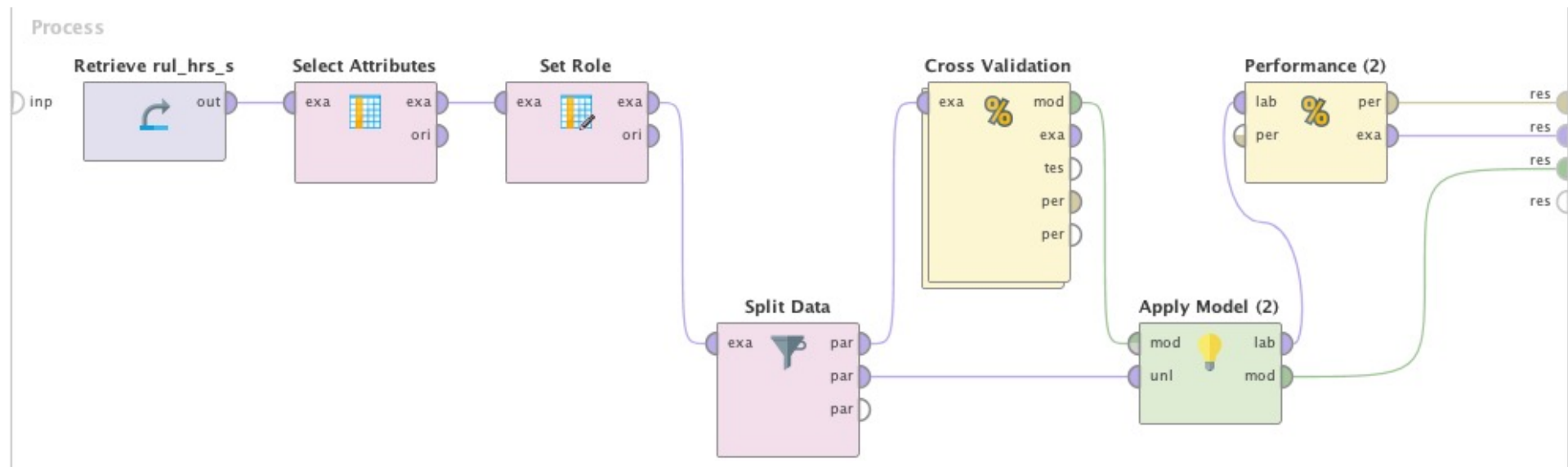
PerformanceVector:

root_mean_squared_error: 93.594 +/- 11.902 (micro average: 94.281 +/- 0.000)

squared_error: 8887.312 +/- 2269.289 (micro average: 8888.936 +/- 23066.583)

correlation: 0.934 +/- 0.023 (micro average: 0.938)

6. Deployment



PerformanceVector

PerformanceVector:
root_mean_squared_error: 92.582 +/- 0.000
squared_error: 8571.455 +/- 19680.990
correlation: 0.943

Assignment

- Data Analyst
- Steel Industry Energy Consumption
- <https://www.kaggle.com/code/bayunova/steel-industry-energy-consumption/data>

Q & A